

Loan.R

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```
library(fastDummies)
library(caTools)
library(car)
```

```
## Loading required package: carData
```

```
set.seed(111)

loan = read.csv("HousingFinance.csv")
View(loan)
summary(loan)
```

```

##          ID          Decision      Build_Selfcon      Tier
## Length:1570      Min.    :0.0000      Length:1570      Min.    :1.000
## Class :character  1st Qu.:1.0000      Class :character  1st Qu.:1.000
## Mode  :character  Median  :1.0000      Mode  :character  Median  :2.000
##                                     Mean    :0.7994      Mean    :1.997
##                                     3rd Qu.:1.0000      3rd Qu.:3.000
##                                     Max.    :1.0000      Max.    :3.000
## Accommodation_Class Marital_Status      Employer_Type      OldEmi_d
## Length:1570      Length:1570      Length:1570      Min.    :0.0000
## Class :character  Class :character  Class :character  1st Qu.:0.0000
## Mode  :character  Mode  :character  Mode  :character  Median  :0.0000
##                                     Mean    :0.1299
##                                     3rd Qu.:0.0000
##                                     Max.    :1.0000
##      BankSave_d      Age      YrsAdd      YrsJob
## Min.    :0.0000      Min.    :22.00      Min.    : 0.000      Min.    : 0.00
## 1st Qu.:0.0000      1st Qu.:31.00      1st Qu.: 2.000      1st Qu.: 4.00
## Median  :0.0000      Median  :37.00      Median  : 5.000      Median  :10.00
## Mean    :0.1197      Mean    :37.76      Mean    : 9.841      Mean    :10.93
## 3rd Qu.:0.0000      3rd Qu.:43.00      3rd Qu.:15.000      3rd Qu.:15.00
## Max.    :1.0000      Max.    :63.00      Max.    :61.000      Max.    :60.00
##      TotInc      MarVal      LoanReq      Term
## Min.    : 7000      Min.    : 59960      Min.    : 76000      Min.    : 15.0
## 1st Qu.: 18773      1st Qu.: 789762      1st Qu.: 400000      1st Qu.:168.0
## Median  : 25127      Median  :1072752      Median  : 600000      Median  :180.0
## Mean    : 29773      Mean    :1177865      Mean    : 593115      Mean    :160.1
## 3rd Qu.: 35238      3rd Qu.:1403509      3rd Qu.: 800000      3rd Qu.:180.0
## Max.    :219858      Max.    :7272727      Max.    :1000000      Max.    :180.0
##      CalcEmi      IIR      IAR      OldEmi
## Min.    : 960.1      Min.    : 4.00      Min.    : 5.00      Min.    : 0.0
## 1st Qu.: 6790.9      1st Qu.:28.00      1st Qu.: 48.99      1st Qu.: 0.0
## Median  : 9234.0      Median  :36.00      Median  : 67.00      Median  : 0.0
## Mean    : 9614.7      Mean    :35.49      Mean    : 64.43      Mean    : 770.1
## 3rd Qu.:12312.0      3rd Qu.:44.00      3rd Qu.: 82.00      3rd Qu.: 0.0
## Max.    :19882.1      Max.    :65.00      Max.    :100.00      Max.    :40000.0
##      LTV      FOIR
## Min.    : 4.00      Min.    : 4.00
## 1st Qu.: 42.00      1st Qu.:30.00
## Median  : 57.00      Median  :38.00
## Mean    : 56.55      Mean    :37.36
## 3rd Qu.: 71.00      3rd Qu.:46.00
## Max.    :222.22      Max.    :73.25

```

```

loan = loan[,-1]
summary(loan)

```

```
##      Decision      Build_Selfcon      Tier      Accommodation_Class
## Min.      :0.0000      Length:1570      Min.      :1.000      Length:1570
## 1st Qu.:1.0000      Class :character      1st Qu.:1.000      Class :character
## Median :1.0000      Mode  :character      Median :2.000      Mode  :character
## Mean    :0.7994      Mean    :1.997
## 3rd Qu.:1.0000      3rd Qu.:3.000
## Max.    :1.0000      Max.    :3.000
## Marital_Status      Employer_Type      OldEmi_d      BankSave_d
## Length:1570      Length:1570      Min.      :0.0000      Min.      :0.0000
## Class :character      Class :character      1st Qu.:0.0000      1st Qu.:0.0000
## Mode  :character      Mode  :character      Median :0.0000      Median :0.0000
##                               Mean    :0.1299      Mean    :0.1197
##                               3rd Qu.:0.0000      3rd Qu.:0.0000
##                               Max.    :1.0000      Max.    :1.0000
##      Age      YrsAdd      YrsJob      TotInc
## Min.      :22.00      Min.      : 0.000      Min.      : 0.00      Min.      : 7000
## 1st Qu.:31.00      1st Qu.: 2.000      1st Qu.: 4.00      1st Qu.: 18773
## Median :37.00      Median : 5.000      Median :10.00      Median : 25127
## Mean    :37.76      Mean    : 9.841      Mean    :10.93      Mean    : 29773
## 3rd Qu.:43.00      3rd Qu.:15.000      3rd Qu.:15.00      3rd Qu.: 35238
## Max.    :63.00      Max.    :61.000      Max.    :60.00      Max.    :219858
##      MarVal      LoanReq      Term      CalcEmi
## Min.      : 59960      Min.      : 76000      Min.      : 15.0      Min.      : 960.1
## 1st Qu.: 789762      1st Qu.: 400000      1st Qu.:168.0      1st Qu.: 6790.9
## Median :1072752      Median : 600000      Median :180.0      Median : 9234.0
## Mean    :1177865      Mean    : 593115      Mean    :160.1      Mean    : 9614.7
## 3rd Qu.:1403509      3rd Qu.: 800000      3rd Qu.:180.0      3rd Qu.:12312.0
## Max.    :7272727      Max.    :1000000      Max.    :180.0      Max.    :19882.1
##      IIR      IAR      OldEmi      LTV
## Min.      : 4.00      Min.      : 5.00      Min.      : 0.0      Min.      : 4.00
## 1st Qu.:28.00      1st Qu.: 48.99      1st Qu.: 0.0      1st Qu.: 42.00
## Median :36.00      Median : 67.00      Median : 0.0      Median : 57.00
## Mean    :35.49      Mean    : 64.43      Mean    : 770.1      Mean    : 56.55
## 3rd Qu.:44.00      3rd Qu.: 82.00      3rd Qu.: 0.0      3rd Qu.: 71.00
## Max.    :65.00      Max.    :100.00      Max.    :40000.0      Max.    :222.22
##      FOIR
## Min.      : 4.00
## 1st Qu.:30.00
## Median :38.00
## Mean    :37.36
## 3rd Qu.:46.00
## Max.    :73.25
```

```
loan = dummy_cols(loan,select_columns = c("Employer_Type","Tier"))
loan$Build_Selfcon = ifelse(loan$Build_Selfcon=="Builder",1,0)
loan$Accommodation_Class = ifelse(loan$Accommodation_Class=="Non_Rented",1,0)
loan$Marital_Status = ifelse(loan$Marital_Status=="Married",1,0)

summary(loan)
```

```
##      Decision      Build_Selfcon      Tier      Accommodation_Class
## Min.      :0.0000      Min.      :0.00000      Min.      :1.000      Min.      :0.0000
```

##	1st Qu.:1.0000	1st Qu.:0.00000	1st Qu.:1.000	1st Qu.:0.0000
##	Median :1.0000	Median :0.00000	Median :2.000	Median :1.0000
##	Mean :0.7994	Mean :0.03758	Mean :1.997	Mean :0.5096
##	3rd Qu.:1.0000	3rd Qu.:0.00000	3rd Qu.:3.000	3rd Qu.:1.0000
##	Max. :1.0000	Max. :1.00000	Max. :3.000	Max. :1.0000
##	Marital_Status	Employer_Type	OldEmi_d	BankSave_d
##	Min. :0.0000	Length:1570	Min. :0.0000	Min. :0.0000
##	1st Qu.:1.0000	Class :character	1st Qu.:0.0000	1st Qu.:0.0000
##	Median :1.0000	Mode :character	Median :0.0000	Median :0.0000
##	Mean :0.9019		Mean :0.1299	Mean :0.1197
##	3rd Qu.:1.0000		3rd Qu.:0.0000	3rd Qu.:0.0000
##	Max. :1.0000		Max. :1.0000	Max. :1.0000
##	Age	YrsAdd	YrsJob	TotInc
##	Min. :22.00	Min. : 0.000	Min. : 0.00	Min. : 7000
##	1st Qu.:31.00	1st Qu.: 2.000	1st Qu.: 4.00	1st Qu.: 18773
##	Median :37.00	Median : 5.000	Median :10.00	Median : 25127
##	Mean :37.76	Mean : 9.841	Mean :10.93	Mean : 29773
##	3rd Qu.:43.00	3rd Qu.:15.000	3rd Qu.:15.00	3rd Qu.: 35238
##	Max. :63.00	Max. :61.000	Max. :60.00	Max. :219858
##	MarVal	LoanReq	Term	CalcEmi
##	Min. : 59960	Min. : 76000	Min. : 15.0	Min. : 960.1
##	1st Qu.: 789762	1st Qu.: 400000	1st Qu.:168.0	1st Qu.: 6790.9
##	Median :1072752	Median : 600000	Median :180.0	Median : 9234.0
##	Mean :1177865	Mean : 593115	Mean :160.1	Mean : 9614.7
##	3rd Qu.:1403509	3rd Qu.: 800000	3rd Qu.:180.0	3rd Qu.:12312.0
##	Max. :7272727	Max. :1000000	Max. :180.0	Max. :19882.1
##	IIR	IAR	OldEmi	LTV
##	Min. : 4.00	Min. : 5.00	Min. : 0.0	Min. : 4.00
##	1st Qu.:28.00	1st Qu.: 48.99	1st Qu.: 0.0	1st Qu.: 42.00
##	Median :36.00	Median : 67.00	Median : 0.0	Median : 57.00
##	Mean :35.49	Mean : 64.43	Mean : 770.1	Mean : 56.55
##	3rd Qu.:44.00	3rd Qu.: 82.00	3rd Qu.: 0.0	3rd Qu.: 71.00
##	Max. :65.00	Max. :100.00	Max. :40000.0	Max. :222.22
##	FOIR	Employer_Type_Business	Employer_Type_Corporate	
##	Min. : 4.00	Min. :0.0000	Min. :0.0000	
##	1st Qu.:30.00	1st Qu.:0.0000	1st Qu.:0.0000	
##	Median :38.00	Median :1.0000	Median :0.0000	
##	Mean :37.36	Mean :0.5089	Mean :0.1841	
##	3rd Qu.:46.00	3rd Qu.:1.0000	3rd Qu.:0.0000	
##	Max. :73.25	Max. :1.0000	Max. :1.0000	
##	Employer_Type_Govt	Employer_Type_Ind/SmallBus	Tier_1	
##	Min. :0.00000	Min. :0.0000	Min. :0.0000	
##	1st Qu.:0.00000	1st Qu.:0.0000	1st Qu.:0.0000	
##	Median :0.00000	Median :0.0000	Median :0.0000	
##	Mean :0.08153	Mean :0.2255	Mean :0.3153	
##	3rd Qu.:0.00000	3rd Qu.:0.0000	3rd Qu.:1.0000	
##	Max. :1.00000	Max. :1.0000	Max. :1.0000	
##	Tier_2	Tier_3		
##	Min. :0.0000	Min. :0.0000		
##	1st Qu.:0.0000	1st Qu.:0.0000		
##	Median :0.0000	Median :0.0000		
##	Mean :0.3726	Mean :0.3121		
##	3rd Qu.:1.0000	3rd Qu.:1.0000		
##	Max. :1.0000	Max. :1.0000		

Correlation

```
cr = cor(loan[sapply(loan,is.numeric)])  
cr
```

##	Decision	Build_Selfcon	Tier
## Decision	1.00000000	0.015367755	-0.068277510
## Build_Selfcon	0.01536776	1.000000000	-0.193712302
## Tier	-0.06827751	-0.193712302	1.000000000
## Accommodation_Class	-0.11609441	-0.013825996	-0.024856273
## Marital_Status	0.01658669	-0.013655787	-0.012140392
## OldEmi_d	0.07062200	0.003324548	-0.012796006
## BankSave_d	0.08190666	0.195332500	-0.008424495
## Age	-0.03462482	0.016104951	-0.055110437
## YrsAdd	-0.09018292	0.032088693	-0.056773575
## YrsJob	-0.07854552	-0.042718486	0.003086885
## TotInc	-0.07897947	-0.125680076	0.134286592
## MarVal	0.12037196	-0.183593321	0.056972176
## LoanReq	-0.03250669	-0.172448245	0.092313847
## Term	0.07347849	0.043128205	-0.078389038
## CalcEmi	-0.05512774	-0.196527100	0.130883884
## IIR	0.01749662	-0.013864674	-0.047413008
## IAR	0.20682078	0.045737240	-0.150800471
## OldEmi	0.01629877	-0.026701707	0.034130416
## LTV	-0.25630467	0.139853358	0.004302393
## FOIR	0.03558534	-0.017063435	-0.043102658
## Employer_Type_Business	-0.11354857	-0.020273382	0.004093095
## Employer_Type_Corporate	0.03276602	0.027131705	0.097358366
## Employer_Type_Govt	0.01558538	-0.058872995	0.015891162
## Employer_Type_Ind/SmallBus	0.09524375	0.037642328	-0.105589913
## Tier_1	-0.00576689	0.233536076	-0.853981137
## Tier_2	0.12293624	-0.131502751	0.003098599
## Tier_3	-0.12249929	-0.096959465	0.853104611
##	Accommodation_Class	Marital_Status	OldEmi_d
## Decision	-0.116094415	0.016586694	0.0706220035
## Build_Selfcon	-0.013825996	-0.013655787	0.0033245481
## Tier	-0.024856273	-0.012140392	-0.0127960056
## Accommodation_Class	1.000000000	0.002019061	0.0077718620
## Marital_Status	0.002019061	1.000000000	-0.0063040347
## OldEmi_d	0.007771862	-0.006304035	1.0000000000
## BankSave_d	0.008648739	0.002907242	0.0033373592
## Age	0.117239098	0.313611186	0.0297164843
## YrsAdd	0.416381349	0.006432943	0.0461872381
## YrsJob	0.156171358	0.176795560	0.0546750661
## TotInc	-0.026239572	0.003881928	0.1782364821
## MarVal	0.052722803	0.009480476	0.0337833856
## LoanReq	-0.140611931	-0.035794283	0.0611517924
## Term	-0.248262860	-0.004814624	-0.0210975636
## CalcEmi	-0.075726954	-0.039263993	0.0636776267
## IIR	-0.018723318	-0.058731131	-0.2297132057
## IAR	-0.132555506	-0.004622967	-0.0439011063
## OldEmi	-0.010362369	0.024441968	0.6426452570
## LTV	-0.143020841	-0.040869091	-0.0004227588

##	FOIR	-0.018426290	-0.050817916	0.2275665697
##	Employer_Type_Business	-0.002889542	0.078703316	0.0840499854
##	Employer_Type_Corporate	-0.033735076	-0.042283619	-0.0466886710
##	Employer_Type_Govt	0.110708258	0.035649045	-0.0320645997
##	Employer_Type_Ind/SmallBus	-0.037751785	-0.078281873	-0.0362528751
##	Tier_1	0.051473411	0.007162935	-0.0257611002
##	Tier_2	-0.058209341	0.006121695	0.0704738846
##	Tier_3	0.009125096	-0.013570599	-0.0477062323
##		BankSave_d	Age	YrsAdd
##	Decision	0.081906661	-0.034624818	-0.090182922
##	Build_Selfcon	0.195332500	0.016104951	0.032088693
##	Tier	-0.008424495	-0.055110437	-0.056773575
##	Accommodation_Class	0.008648739	0.117239098	0.416381349
##	Marital_Status	0.002907242	0.313611186	0.006432943
##	OldEmi_d	0.003337359	0.029716484	0.046187238
##	BankSave_d	1.000000000	0.043162308	0.028071941
##	Age	0.043162308	1.000000000	0.191919780
##	YrsAdd	0.028071941	0.191919780	1.000000000
##	YrsJob	-0.001402737	0.450935718	0.217080289
##	TotInc	-0.013257096	-0.027840811	-0.001213283
##	MarVal	-0.034902147	-0.002120545	-0.031460879
##	LoanReq	-0.029308103	-0.117839399	-0.062359502
##	Term	0.010322635	-0.291982686	-0.215392648
##	CalcEmi	-0.038443686	-0.045560393	-0.001000594
##	IIR	-0.009264779	-0.024179361	-0.022867832
##	IAR	0.027608747	0.013271056	-0.097476316
##	OldEmi	-0.018118736	0.025560762	0.006167054
##	LTV	0.013264344	-0.076890836	-0.002519474
##	FOIR	-0.011985095	-0.002235816	-0.007058284
##	Employer_Type_Business	-0.006578886	-0.035091371	0.031641292
##	Employer_Type_Corporate	0.098175323	-0.053034381	-0.018818595
##	Employer_Type_Govt	-0.016685836	0.197467951	0.082051570
##	Employer_Type_Ind/SmallBus	-0.072248946	-0.038143386	-0.074128574
##	Tier_1	0.117070882	0.062456714	0.071276460
##	Tier_2	-0.211202650	-0.029756099	-0.043982596
##	Tier_3	0.102992783	-0.031579045	-0.025577997
##		TotInc	MarVal	LoanReq
##	Decision	-0.078979465	0.120371960	-0.03250669
##	Build_Selfcon	-0.125680076	-0.183593321	-0.17244825
##	Tier	0.134286592	0.056972176	0.09231385
##	Accommodation_Class	-0.026239572	0.052722803	-0.14061193
##	Marital_Status	0.003881928	0.009480476	-0.03579428
##	OldEmi_d	0.178236482	0.033783386	0.06115179
##	BankSave_d	-0.013257096	-0.034902147	-0.02930810
##	Age	-0.027840811	-0.002120545	-0.11783940
##	YrsAdd	-0.001213283	-0.031460879	-0.06235950
##	YrsJob	0.094424005	0.042002503	0.01355231
##	TotInc	1.000000000	0.320257795	0.53285709
##	MarVal	0.320257795	1.000000000	0.48914012
##	LoanReq	0.532857089	0.489140118	1.000000000
##	Term	-0.003172789	0.063128743	0.34548797
##	CalcEmi	0.569357675	0.503607581	0.95059072
##	IIR	-0.500009999	0.068274230	0.23246148
##	IAR	-0.539252641	-0.024375016	-0.03079678

##	OldEmi	0.313574063	0.091733864	0.12347256	-0.003465678
##	LTV	0.072857138	-0.504600573	0.21965988	0.181792577
##	FOIR	-0.384745482	0.091822518	0.25568949	0.079283855
##	Employer_Type_Business	0.279500606	0.102707542	0.14868102	0.048723994
##	Employer_Type_Corporate	-0.059748899	0.019891326	0.06692981	-0.032600226
##	Employer_Type_Govt	-0.087276752	0.015371525	-0.01590563	-0.093784364
##	Employer_Type_Ind/SmallBus	-0.221800663	-0.151379009	-0.22951758	0.033356653
##	Tier_1	-0.098020908	-0.063510194	-0.02956177	0.059072198
##	Tier_2	-0.031598317	0.028730918	-0.09441298	0.014883884
##	Tier_3	0.131263790	0.033705193	0.12816186	-0.074766332
##		CalcEmi	IIR	IAR	OldEmi
##	Decision	-0.055127739	0.017496618	0.206820781	0.016298767
##	Build_Selfcon	-0.196527100	-0.013864674	0.045737240	-0.026701707
##	Tier	0.130883884	-0.047413008	-0.150800471	0.034130416
##	Accommodation_Class	-0.075726954	-0.018723318	-0.132555506	-0.010362369
##	Marital_Status	-0.039263993	-0.058731131	-0.004622967	0.024441968
##	OldEmi_d	0.063677627	-0.229713206	-0.043901106	0.642645257
##	BankSave_d	-0.038443686	-0.009264779	0.027608747	-0.018118736
##	Age	-0.045560393	-0.024179361	0.013271056	0.025560762
##	YrsAdd	-0.001000594	-0.022867832	-0.097476316	0.006167054
##	YrsJob	0.064066450	-0.101865588	-0.089402937	0.028538070
##	TotInc	0.569357675	-0.500009999	-0.539252641	0.313574063
##	MarVal	0.503607581	0.068274230	-0.024375016	0.091733864
##	LoanReq	0.950590722	0.232461484	-0.030796778	0.123472563
##	Term	0.070578663	0.094686310	0.103456576	-0.003465678
##	CalcEmi	1.000000000	0.231589512	-0.056170526	0.126808502
##	IIR	0.231589512	1.000000000	0.710751176	-0.254548944
##	IAR	-0.056170526	0.710751176	1.000000000	-0.105706561
##	OldEmi	0.126808502	-0.254548944	-0.105706561	1.000000000
##	LTV	0.182268713	0.087193996	-0.036979699	-0.002119156
##	FOIR	0.255307763	0.853041382	0.670587592	0.223729945
##	Employer_Type_Business	0.140980526	-0.277417358	-0.222904077	0.074299067
##	Employer_Type_Corporate	0.082620601	0.177665454	0.091053085	-0.008847985
##	Employer_Type_Govt	0.003116611	0.141702130	0.085192270	-0.019734154
##	Employer_Type_Ind/SmallBus	-0.247312949	0.074317132	0.126429728	-0.067754717
##	Tier_1	-0.063302732	0.064169359	0.137756858	-0.044106362
##	Tier_2	-0.092750381	-0.045657677	-0.017718834	0.028857333
##	Tier_3	0.160261047	-0.016703361	-0.119647705	0.014115899
##		LTV	FOIR	Employer_Type_Business	
##	Decision	-0.2563046698	0.035585339		-0.1135485670
##	Build_Selfcon	0.1398533578	-0.017063435		-0.0202733825
##	Tier	0.0043023929	-0.043102658		0.0040930945
##	Accommodation_Class	-0.1430208410	-0.018426290		-0.0028895424
##	Marital_Status	-0.0408690911	-0.050817916		0.0787033165
##	OldEmi_d	-0.0004227588	0.227566570		0.0840499854
##	BankSave_d	0.0132643443	-0.011985095		-0.0065788860
##	Age	-0.0768908361	-0.002235816		-0.0350913708
##	YrsAdd	-0.0025194742	-0.007058284		0.0316412916
##	YrsJob	-0.0220807823	-0.077001002		0.3401627374
##	TotInc	0.0728571381	-0.384745482		0.2795006064
##	MarVal	-0.5046005730	0.091822518		0.1027075416
##	LoanReq	0.2196598820	0.255689485		0.1486810242
##	Term	0.1817925767	0.079283855		0.0487239944
##	CalcEmi	0.1822687125	0.255307763		0.1409805256

## IIR	0.0871939956	0.853041382	-0.2774173579
## IAR	-0.0369796985	0.670587592	-0.2229040773
## OldEmi	-0.0021191563	0.223729945	0.0742990668
## LTV	1.0000000000	0.075390624	0.0645272642
## FOIR	0.0753906235	1.000000000	-0.2404418755
## Employer_Type_Business	0.0645272642	-0.240441875	1.0000000000
## Employer_Type_Corporate	0.0177628639	0.169606423	-0.4835268076
## Employer_Type_Govt	-0.0637915881	0.131535887	-0.3032973077
## Employer_Type_Ind/SmallBus	-0.0518936341	0.044214927	-0.5492638847
## Tier_1	0.0577574785	0.050705471	0.0002357905
## Tier_2	-0.1180548558	-0.026842075	-0.0071585063
## Tier_3	0.0652715711	-0.022836110	0.0072333513
##	Employer_Type_Corporate	Employer_Type_Govt	
## Decision	0.032766022	0.015585378	
## Build_Selfcon	0.027131705	-0.058872995	
## Tier	0.097358366	0.015891162	
## Accommodation_Class	-0.033735076	0.110708258	
## Marital_Status	-0.042283619	0.035649045	
## OldEmi_d	-0.046688671	-0.032064600	
## BankSave_d	0.098175323	-0.016685836	
## Age	-0.053034381	0.197467951	
## YrsAdd	-0.018818595	0.082051570	
## YrsJob	-0.289767588	0.226842158	
## TotInc	-0.059748899	-0.087276752	
## MarVal	0.019891326	0.015371525	
## LoanReq	0.066929810	-0.015905632	
## Term	-0.032600226	-0.093784364	
## CalcEmi	0.082620601	0.003116611	
## IIR	0.177665454	0.141702130	
## IAR	0.091053085	0.085192270	
## OldEmi	-0.008847985	-0.019734154	
## LTV	0.017762864	-0.063791588	
## FOIR	0.169606423	0.131535887	
## Employer_Type_Business	-0.483526808	-0.303297308	
## Employer_Type_Corporate	1.000000000	-0.141513122	
## Employer_Type_Govt	-0.141513122	1.000000000	
## Employer_Type_Ind/SmallBus	-0.256276745	-0.160752301	
## Tier_1	-0.014565930	0.033280501	
## Tier_2	-0.131498107	-0.089996256	
## Tier_3	0.151822406	0.060537377	
##	Employer_Type_Ind/SmallBus	Tier_1	
## Decision	0.095243747	-0.0057668905	
## Build_Selfcon	0.037642328	0.2335360763	
## Tier	-0.105589913	-0.8539811375	
## Accommodation_Class	-0.037751785	0.0514734109	
## Marital_Status	-0.078281873	0.0071629350	
## OldEmi_d	-0.036252875	-0.0257611002	
## BankSave_d	-0.072248946	0.1170708818	
## Age	-0.038143386	0.0624567136	
## YrsAdd	-0.074128574	0.0712764596	
## YrsJob	-0.286746065	0.0273014460	
## TotInc	-0.221800663	-0.0980209081	
## MarVal	-0.151379009	-0.0635101943	
## LoanReq	-0.229517584	-0.0295617707	

## Term		0.033356653	0.0590721979
## CalcEmi		-0.247312949	-0.0633027318
## IIR		0.074317132	0.0641693587
## IAR		0.126429728	0.1377568583
## OldEmi		-0.067754717	-0.0441063624
## LTV		-0.051893634	0.0577574785
## FOIR		0.044214927	0.0507054709
## Employer_Type_Business		-0.549263885	0.0002357905
## Employer_Type_Corporate		-0.256276745	-0.0145659298
## Employer_Type_Govt		-0.160752301	0.0332805015
## Employer_Type_Ind/SmallBus		1.000000000	-0.0085666023
## Tier_1		-0.008566602	1.0000000000
## Tier_2		0.189442509	-0.5229476132
## Tier_3		-0.189090136	-0.4570719076
##	Tier_2	Tier_3	
## Decision	0.122936235	-0.122499288	
## Build_Selfcon	-0.131502751	-0.096959465	
## Tier	0.003098599	0.853104611	
## Accommodation_Class	-0.058209341	0.009125096	
## Marital_Status	0.006121695	-0.013570599	
## OldEmi_d	0.070473885	-0.047706232	
## BankSave_d	-0.211202650	0.102992783	
## Age	-0.029756099	-0.031579045	
## YrsAdd	-0.043982596	-0.025577997	
## YrsJob	-0.057528808	0.032653645	
## TotInc	-0.031598317	0.131263790	
## MarVal	0.028730918	0.033705193	
## LoanReq	-0.094412979	0.128161865	
## Term	0.014883884	-0.074766332	
## CalcEmi	-0.092750381	0.160261047	
## IIR	-0.045657677	-0.016703361	
## IAR	-0.017718834	-0.119647705	
## OldEmi	0.028857333	0.014115899	
## LTV	-0.118054856	0.065271571	
## FOIR	-0.026842075	-0.022836110	
## Employer_Type_Business	-0.007158506	0.007233351	
## Employer_Type_Corporate	-0.131498107	0.151822406	
## Employer_Type_Govt	-0.089996256	0.060537377	
## Employer_Type_Ind/SmallBus	0.189442509	-0.189090136	
## Tier_1	-0.522947613	-0.457071908	
## Tier_2	1.000000000	-0.519093965	
## Tier_3	-0.519093965	1.000000000	

```
write.csv(cr,"cor_loan.csv")
```

```
## train test split ##
split = sample.split(loan$Decision,SplitRatio = 0.8)
training_data = subset(loan,split == "TRUE")
testing_data = subset(loan,split == "FALSE")
View(training_data)
View(testing_data)

### MODEL ###
modell = glm(Decision ~ IAR,data = training_data,family=binomial)
summary(modell) ## Residual deviance: 1194.5, AIC: 1198.5 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.248398   0.209458  -1.186   0.236
## IAR          0.026878   0.003424   7.850 4.15e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1194.5  on 1254  degrees of freedom
## AIC: 1198.5
##
## Number of Fisher Scoring iterations: 4
```

```
model2 = glm(Decision ~ IAR+MarVal,data = training_data,family=binomial)
summary(model2) ## Residual deviance: 1179.4, AIC: 1185.4 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal, family = binomial, data = training_data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.082e-01  2.597e-01  -3.112 0.001856 **
## IAR          2.686e-02  3.426e-03   7.840 4.51e-15 ***
## MarVal       4.995e-07  1.385e-07   3.607 0.000309 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1179.4  on 1253  degrees of freedom
## AIC: 1185.4
##
## Number of Fisher Scoring iterations: 4
```

```
vif(model2) ## no correlation
```

```
##      IAR      MarVal
## 1.000286 1.000286
```

```
model3 = glm(Decision ~ IAR+MarVal+Term,data = training_data,family=binomial)
summary(model3) ## Residual deviance: 1176.5, AIC: 1184.5 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term, family = binomial,
##      data = training_data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.234e+00  3.597e-01  -3.430 0.000603 ***
## IAR          2.605e-02  3.459e-03   7.532 5e-14 ***
## MarVal       4.703e-07  1.374e-07   3.423 0.000619 ***
## Term         3.198e-03  1.869e-03   1.711 0.086991 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1176.5  on 1252  degrees of freedom
## AIC: 1184.5
##
## Number of Fisher Scoring iterations: 4
```

```
vif(model3)  ## no correlation
```

```
##      IAR   MarVal    Term  
## 1.016580 1.014285 1.029941
```

```
model4 = glm(Decision ~ IAR+MarVal+Term+FOIR,data = training_data,family=binomial)  
summary(model4) ## Residual deviance: 1128.7, AIC: 1138.7 ##
```

```
##  
## Call:  
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR, family = binomial,  
##      data = training_data)  
##  
## Coefficients:  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept) -6.236e-01  3.769e-01  -1.655   0.0980 .  
## IAR          5.365e-02  5.758e-03   9.318  < 2e-16 ***  
## MarVal       6.895e-07  1.547e-07   4.457 8.31e-06 ***  
## Term         3.495e-03  1.898e-03   1.841  0.0656 .  
## FOIR        -6.909e-02  1.038e-02  -6.656 2.81e-11 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## (Dispersion parameter for binomial family taken to be 1)  
##  
##      Null deviance: 1259.2  on 1255  degrees of freedom  
## Residual deviance: 1128.7  on 1251  degrees of freedom  
## AIC: 1138.7  
##  
## Number of Fisher Scoring iterations: 5
```

```
vif(model4)  ## modrate correlation
```

```
##      IAR   MarVal    Term    FOIR  
## 2.478419 1.082533 1.036207 2.541034
```

```
model5 = glm(Decision ~ IAR+MarVal+Term+FOIR+IIR,data = training_data,family=binomial)  
summary(model5) ## Residual deviance: 1086.9, AIC: 1098.9 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + IIR, family = binomial,
##      data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -4.844e-01  3.834e-01  -1.264   0.2064
## IAR          7.637e-02  7.425e-03  10.285 < 2e-16 ***
## MarVal       7.914e-07  1.638e-07   4.830 1.36e-06 ***
## Term         3.453e-03  1.950e-03   1.771  0.0766 .
## FOIR        -1.805e-02  1.529e-02  -1.181  0.2377
## IIR         -9.828e-02  1.697e-02  -5.791 7.01e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1086.9  on 1250  degrees of freedom
## AIC: 1098.9
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model5)  ## IAR    MarVal    Term    FOIR    IIR
```

```
##      IAR    MarVal    Term    FOIR    IIR
## 3.881923 1.099954 1.041943 5.277081 6.503907
```

```
# 3.881923 1.099954 1.041943 5.277081 6.503907
```

```
model6 = glm(Decision ~ IAR+MarVal+Term+FOIR+OldEmi,data = training_data,family=binomial)
summary(model6) ## Residual deviance: 1104.6, AIC: 1116.6 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + OldEmi,
##      family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -5.013e-01  3.806e-01  -1.317   0.1878
## IAR          6.951e-02  7.060e-03   9.846 < 2e-16 ***
## MarVal       7.160e-07  1.586e-07   4.516 6.31e-06 ***
## Term         3.335e-03  1.925e-03   1.732  0.0832 .
## FOIR        -1.005e-01  1.279e-02  -7.858 3.91e-15 ***
## OldEmi       1.461e-04  3.690e-05   3.961 7.48e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1104.6  on 1250  degrees of freedom
## AIC: 1116.6
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model6) ##### modrate correlation
```

```
##      IAR      MarVal      Term      FOIR      OldEmi
## 3.565964 1.078749 1.039051 3.782421 1.308034
```

```
model7 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV,data = training_data,family=binomial)
summary(model7) ## Residual deviance: 1084.5, AIC: 1096.5 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV, family = binomial,
##      data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  9.257e-01  4.655e-01   1.989 0.046756 *
## IAR          4.944e-02  5.912e-03   8.363 < 2e-16 ***
## MarVal       1.289e-08  1.730e-07   0.075 0.940607
## Term        8.154e-03  2.119e-03   3.848 0.000119 ***
## FOIR       -5.678e-02  1.076e-02  -5.277 1.31e-07 ***
## LTV        -2.899e-02  4.872e-03  -5.951 2.67e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1084.5  on 1250  degrees of freedom
## AIC: 1096.5
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model7) ##### modrate correlation
```

```
##      IAR    MarVal    Term    FOIR    LTV
## 2.477608 1.551874 1.189898 2.559574 1.613876
```

```
model8 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd,data = training_data,famil
y=binomial)
summary(model8) ## Residual deviance: 1082.4, AIC: 1096.4 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd,
##      family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.127e+00  4.874e-01   2.312 0.020750 *
## IAR          4.855e-02  5.943e-03   8.169 3.11e-16 ***
## MarVal       9.564e-09  1.737e-07   0.055 0.956094
## Term        7.537e-03  2.168e-03   3.476 0.000509 ***
## FOIR        -5.506e-02  1.082e-02  -5.089 3.61e-07 ***
## LTV         -2.901e-02  4.898e-03  -5.922 3.18e-09 ***
## YrsAdd      -1.002e-02  6.973e-03  -1.437 0.150631
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1082.4  on 1249  degrees of freedom
## AIC: 1096.4
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model8) ##### modrate correlation
```

```
##      IAR    MarVal    Term    FOIR    LTV    YrsAdd
## 2.495166 1.553881 1.241496 2.585766 1.617169 1.061084
```

```
model9 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc,data = training_data,family=binomial)
summary(model9) ## Residual deviance: 1081.6, AIC: 1097.6 ##
```



```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  9.456e-01  5.229e-01   1.808 0.070566 .
## IAR          5.088e-02  6.481e-03   7.852 4.11e-15 ***
## MarVal      -9.234e-08  1.995e-07  -0.463 0.643432
## Term         7.847e-03  2.191e-03   3.581 0.000343 ***
## FOIR        -5.373e-02  1.092e-02  -4.921 8.63e-07 ***
## LTV         -3.103e-02  5.417e-03  -5.728 1.01e-08 ***
## YrsAdd       -9.499e-03  7.001e-03  -1.357 0.174843
## TotInc       5.515e-06  5.908e-06   0.933 0.350572
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1081.6  on 1248  degrees of freedom
## AIC: 1097.6
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model9) ##### modrate correlation
```

```
##      IAR    MarVal    Term    FOIR    LTV    YrsAdd    TotInc
## 2.927114 2.170394 1.270468 2.611668 1.949006 1.067293 2.007441
```

```
model10 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob,data = trai
ning_data,family=binomial)
summary(model10) ## Residual deviance: 1078.8, AIC: 1096.8 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.151e+00  5.396e-01   2.133 0.032965 *
## IAR          5.080e-02  6.483e-03   7.837 4.63e-15 ***
## MarVal      -8.845e-08  2.002e-07  -0.442 0.658596
## Term         7.492e-03  2.210e-03   3.390 0.000698 ***
## FOIR        -5.371e-02  1.092e-02  -4.919 8.71e-07 ***
## LTV         -3.111e-02  5.441e-03  -5.717 1.09e-08 ***
## YrsAdd       -7.109e-03  7.170e-03  -0.992 0.321437
## TotInc       5.872e-06  5.925e-06   0.991 0.321686
## YrsJob      -1.616e-02  9.651e-03  -1.674 0.094073 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1078.8  on 1247  degrees of freedom
## AIC: 1096.8
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model10) ##### modrate correlation
```

```
##      IAR  MarVal  Term  FOIR  LTV  YrsAdd  TotInc  YrsJob
## 2.909938 2.178902 1.280222 2.597767 1.946848 1.110775 2.005620 1.069296
```

```
model11 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+CalcEmi,dat
a = training_data,family=binomial)
summary(model11) ## Residual deviance: 1077.5, AIC: 1097.5 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + CalcEmi, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  9.239e-01  5.767e-01   1.602   0.109
## IAR          5.148e-02  6.551e-03   7.858 3.91e-15 ***
## MarVal       1.067e-07  2.805e-07   0.380   0.704
## Term         6.997e-03  2.264e-03   3.091   0.002 **
## FOIR        -4.983e-02  1.153e-02  -4.321 1.56e-05 ***
## LTV         -2.721e-02  6.337e-03  -4.294 1.76e-05 ***
## YrsAdd       -6.830e-03  7.168e-03  -0.953   0.341
## TotInc       9.893e-06  7.071e-06   1.399   0.162
## YrsJob      -1.568e-02  9.652e-03  -1.625   0.104
## CalcEmi     -4.630e-05  4.152e-05  -1.115   0.265
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1077.5  on 1246  degrees of freedom
## AIC: 1097.5
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model11) ##### modrate correlation
```

```
##      IAR    MarVal    Term    FOIR    LTV    YrsAdd    TotInc    YrsJob
## 2.946897 3.836326 1.339350 2.876215 2.719302 1.110514 2.748497 1.069665
## CalcEmi
## 3.824597
```

```
model12 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Age,data =
training_data,family=binomial)
summary(model12) ## Residual deviance: 1078.4, AIC: 1098.4 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Age, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  8.868e-01  6.746e-01   1.315 0.188669
## IAR          5.074e-02  6.486e-03   7.823 5.15e-15 ***
## MarVal      -8.875e-08  2.005e-07  -0.443 0.658042
## Term         7.861e-03  2.283e-03   3.444 0.000574 ***
## FOIR        -5.394e-02  1.093e-02  -4.935 8.01e-07 ***
## LTV         -3.113e-02  5.440e-03  -5.722 1.05e-08 ***
## YrsAdd      -7.437e-03  7.198e-03  -1.033 0.301511
## TotInc       5.904e-06  5.934e-06   0.995 0.319776
## YrsJob      -1.928e-02  1.082e-02  -1.782 0.074748 .
## Age         6.800e-03  1.048e-02   0.649 0.516467
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1078.4  on 1246  degrees of freedom
## AIC: 1098.4
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model12) ##### modrate correlation
```

```
##      IAR  MarVal  Term  FOIR  LTV  YrsAdd  TotInc  YrsJob
## 2.909951 2.179152 1.366764 2.598804 1.948426 1.116623 2.007083 1.339581
##      Age
## 1.426339
```

```
model13 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+LoanReq, data = training_data, family=binomial)
summary(model13) ## Residual deviance: 1077.4, AIC: 1097.4 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + LoanReq, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  7.097e-01  6.563e-01   1.081 0.279553
## IAR          5.142e-02  6.544e-03   7.857 3.92e-15 ***
## MarVal       1.135e-07  2.800e-07   0.405 0.685248
## Term         8.474e-03  2.369e-03   3.577 0.000348 ***
## FOIR        -5.002e-02  1.144e-02  -4.374 1.22e-05 ***
## LTV         -2.716e-02  6.267e-03  -4.334 1.46e-05 ***
## YrsAdd       -6.863e-03  7.159e-03  -0.959 0.337740
## TotInc       9.800e-06  6.943e-06   1.411 0.158142
## YrsJob       -1.577e-02  9.637e-03  -1.636 0.101812
## LoanReq      -7.784e-07  6.634e-07  -1.173 0.240684
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1077.4  on 1246  degrees of freedom
## AIC: 1097.4
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model13) ##### IAR   MarVal   Term   FOIR   LTV   YrsAdd   TotInc   Yr
sJob   LoanReq
```

```
##      IAR   MarVal   Term   FOIR   LTV   YrsAdd   TotInc   YrsJob
## 2.944150 3.796767 1.466922 2.831063 2.664446 1.110306 2.648439 1.068851
##   LoanReq
## 4.290407
```

```
# 2.944150 3.796767 1.466922 2.831063 2.664446 1.110306 2.648439 1.0
68851 4.290407
```

```
model14 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2,data
= training_data,family=binomial)
summary(model14) ## Residual deviance: 1065.1, AIC: 1085.1 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  8.068e-01  5.490e-01   1.470 0.141676
## IAR          5.253e-02  6.542e-03   8.029 9.84e-16 ***
## MarVal      -8.062e-08  2.023e-07  -0.399 0.690217
## Term         7.309e-03  2.221e-03   3.290 0.001001 **
## FOIR        -5.491e-02  1.098e-02  -5.002 5.68e-07 ***
## LTV         -2.990e-02  5.441e-03  -5.496 3.88e-08 ***
## YrsAdd      -7.208e-03  7.230e-03  -0.997 0.318835
## TotInc       6.419e-06  5.968e-06   1.076 0.282144
## YrsJob      -1.450e-02  9.729e-03  -1.491 0.136076
## Tier_2       6.143e-01  1.700e-01   3.613 0.000302 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1065.1  on 1246  degrees of freedom
## AIC: 1085.1
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model14) ##### modrate correlation
```

```
##      IAR    MarVal    Term    FOIR    LTV    YrsAdd    TotInc    YrsJob
## 2.955577 2.160568 1.279404 2.624862 1.938864 1.108545 2.007332 1.074004
## Tier_2
## 1.014694
```

```
colnames(training_data)[25] <- "Employer_Type_Ind_SmallBus"
colnames(testing_data)[25] <- "Employer_Type_Ind_SmallBus"

model15 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Employer_Type_Ind_SmallBus,data = training_data,family=binomial)
summary(model15) ## Residual deviance: 1063.3, AIC: 1085.3 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + Employer_Type_Ind_SmallBus, family = binomial,
##      data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    6.699e-01  5.585e-01   1.200 0.230329
## IAR            5.217e-02  6.546e-03   7.970 1.59e-15 ***
## MarVal        -5.492e-08  2.043e-07  -0.269 0.788027
## Term          7.211e-03  2.224e-03   3.242 0.001186 **
## FOIR          -5.464e-02  1.099e-02  -4.972 6.64e-07 ***
## LTV           -2.933e-02  5.440e-03  -5.391 7.01e-08 ***
## YrsAdd        -6.963e-03  7.237e-03  -0.962 0.335953
## TotInc         7.116e-06  6.014e-06   1.183 0.236740
## YrsJob        -1.166e-02  9.962e-03  -1.170 0.241933
## Tier_2         5.798e-01  1.720e-01   3.371 0.000749 ***
## Employer_Type_Ind_SmallBus  2.913e-01  2.201e-01   1.323 0.185764
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1063.3  on 1245  degrees of freedom
## AIC: 1085.3
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model15) ##### modrate correlation
```

```
##              IAR              MarVal
##      2.965754          2.180403
##              Term              FOIR
##      1.280015          2.634539
##              LTV              YrsAdd
##      1.946373          1.110240
##              TotInc              YrsJob
##      2.019021          1.120194
##              Tier_2 Employer_Type_Ind_SmallBus
##      1.035089          1.119465
```

```
model16 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d,data = training_data,family=binomial)
summary(model16) ## Residual deviance: 1043.6, AIC: 1065.6 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d, family = binomial,
##      data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  6.692e-01  5.559e-01   1.204  0.22868
## IAR          5.235e-02  6.643e-03   7.880 3.28e-15 ***
## MarVal      -2.054e-08  2.085e-07  -0.099  0.92153
## Term        6.925e-03  2.260e-03   3.064  0.00219 **
## FOIR       -5.469e-02  1.113e-02  -4.914 8.94e-07 ***
## LTV        -2.946e-02  5.517e-03  -5.339 9.35e-08 ***
## YrsAdd      -7.596e-03  7.307e-03  -1.040  0.29857
## TotInc       5.507e-06  6.055e-06   0.910  0.36307
## YrsJob     -1.630e-02  9.872e-03  -1.651  0.09878 .
## Tier_2       7.404e-01  1.724e-01   4.294 1.75e-05 ***
## BankSave_d   1.306e+00  3.173e-01   4.115 3.87e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1043.6  on 1245  degrees of freedom
## AIC: 1065.6
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model16) ##### modrate correlation
```

```
##          IAR      MarVal      Term      FOIR      LTV      YrsAdd      TotInc
##  2.969142  2.185225  1.296130  2.660321  1.941671  1.112091  1.995181
##    YrsJob    Tier_2 BankSave_d
##  1.081443  1.030899  1.028970
```

```
model17 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d+OldEmi_d,data = training_data,family=binomial)
summary(model17) ## Residual deviance: 1007.5, AIC: 1031.5 ##
```



```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d, family = binomial,
##      data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.109e+00  5.743e-01   1.932   0.0534 .
## IAR          6.546e-02  7.508e-03   8.718 < 2e-16 ***
## MarVal       3.204e-07  2.421e-07   1.323   0.1857
## Term        5.743e-03  2.321e-03   2.475   0.0133 *
## FOIR       -9.556e-02  1.378e-02  -6.932  4.14e-12 ***
## LTV        -2.377e-02  5.321e-03  -4.468  7.91e-06 ***
## YrsAdd     -8.485e-03  7.425e-03  -1.143   0.2532
## TotInc     -5.640e-06  6.484e-06  -0.870   0.3844
## YrsJob     -2.031e-02  1.006e-02  -2.018   0.0436 *
## Tier_2      7.016e-01  1.754e-01   3.999  6.36e-05 ***
## BankSave_d  1.312e+00  3.222e-01   4.070  4.69e-05 ***
## OldEmi_d    1.785e+00  3.289e-01   5.427  5.74e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1007.5  on 1244  degrees of freedom
## AIC: 1031.5
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model17) ##### IAR      MarVal      Term      FOIR      LTV      YrsAdd      T
otInc      YrsJob      Tier_2      BankSave_d      OldEmi_d
```

```
##      IAR      MarVal      Term      FOIR      LTV      YrsAdd      TotInc
##  3.701032  2.344866  1.302588  4.012666  1.894695  1.116851  2.273076
##      YrsJob      Tier_2 BankSave_d      OldEmi_d
##  1.101772  1.032997  1.045017  1.394694
```

```
## 3.701032  2.344866  1.302588  4.012666  1.894695  1.116851
2.273076  1.101772  1.032997  1.045017  1.394694
```

```
model18 = glm(Decision ~ IAR+MarVal+Term+LTV+YrsAdd+TotInc+YrsJob+Tier_2+BankSave_
d+OldEmi_d,data = training_data,family=binomial)
summary(model18) ## Residual deviance: 1060.4, AIC: 1082.4 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d, family = binomial,
##      data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.506e-01  5.498e-01   0.820  0.41247
## IAR          3.073e-02  4.806e-03   6.394 1.62e-10 ***
## MarVal      -2.428e-07  1.779e-07  -1.364  0.17246
## Term        7.134e-03  2.238e-03   3.188  0.00143 **
## LTV        -3.535e-02  5.447e-03  -6.490 8.57e-11 ***
## YrsAdd     -1.261e-02  7.249e-03  -1.740  0.08182 .
## TotInc      6.789e-06  5.937e-06   1.143  0.25286
## YrsJob     -1.750e-02  9.895e-03  -1.769  0.07695 .
## Tier_2      6.976e-01  1.718e-01   4.061 4.88e-05 ***
## BankSave_d  1.330e+00  3.174e-01   4.190 2.79e-05 ***
## OldEmi_d    7.348e-01  2.733e-01   2.689  0.00716 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1060.4  on 1245  degrees of freedom
## AIC: 1082.4
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model18) ## no correlation
```

```
##          IAR      MarVal      Term      LTV      YrsAdd      TotInc      YrsJob
##  1.670803  1.982679  1.266674  1.837061  1.110024  1.997936  1.088363
##      Tier_2 BankSave_d  OldEmi_d
##  1.032082  1.032662  1.037687
```

```
model19 = glm(Decision ~ MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+BankSave
_d+OldEmi_d,data = training_data,family=binomial)
summary(model19) ## Residual deviance: 1102.4, AIC: 1124.4 ##
```

```
##
## Call:
## glm(formula = Decision ~ MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d, family = binomial,
##      data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.602e+00  5.550e-01   4.688 2.76e-06 ***
## MarVal       2.988e-07  2.316e-07   1.290 0.196943
## Term        6.469e-03  2.209e-03   2.928 0.003407 **
## FOIR       -1.167e-02  9.463e-03  -1.233 0.217436
## LTV        -2.441e-02  5.162e-03  -4.729 2.25e-06 ***
## YrsAdd     -1.792e-02  7.040e-03  -2.545 0.010925 *
## TotInc     -2.323e-05  6.628e-06  -3.505 0.000457 ***
## YrsJob     -1.938e-02  9.751e-03  -1.987 0.046903 *
## Tier_2      6.135e-01  1.690e-01   3.630 0.000284 ***
## BankSave_d  1.368e+00  3.091e-01   4.426 9.60e-06 ***
## OldEmi_d    1.041e+00  2.960e-01   3.516 0.000439 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1102.4  on 1245  degrees of freedom
## AIC: 1124.4
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model19) ##### modrate correlation
```

```
##      MarVal      Term      FOIR      LTV      YrsAdd      TotInc      YrsJob
##  2.464196  1.271921  1.888557  1.884387  1.103697  2.353008  1.095010
##      Tier_2 BankSave_d  OldEmi_d
##  1.032994  1.036870  1.244884
```

```
model20 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d+OldEmi_d+Employer_Type_Corporate,data = training_data,family=binomial)
summary(model20) ## Residual deviance: 1006.1, AIC: 1032.1 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d + Employer_Type_Corporate,
##      family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.024e+00  5.784e-01   1.770  0.07666 .
## IAR              6.593e-02  7.520e-03   8.767 < 2e-16 ***
## MarVal          3.124e-07  2.420e-07   1.291  0.19669
## Term            6.036e-03  2.334e-03   2.586  0.00972 **
## FOIR           -9.750e-02  1.390e-02  -7.014 2.32e-12 ***
## LTV            -2.394e-02  5.330e-03  -4.493 7.03e-06 ***
## YrsAdd         -8.681e-03  7.427e-03  -1.169  0.24242
## TotInc         -5.489e-06  6.475e-06  -0.848  0.39661
## YrsJob         -1.623e-02  1.067e-02  -1.521  0.12830
## Tier_2          7.227e-01  1.765e-01   4.094 4.24e-05 ***
## BankSave_d      1.285e+00  3.230e-01   3.979 6.93e-05 ***
## OldEmi_d        1.802e+00  3.294e-01   5.470 4.51e-08 ***
## Employer_Type_Corporate 2.605e-01  2.256e-01   1.155  0.24814
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1006.1  on 1243  degrees of freedom
## AIC: 1032.1
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model20) ##### more than modrate correlation
```

##	IAR	MarVal	Term
##	3.708702	2.341095	1.320473
##	FOIR	LTV	YrsAdd
##	4.067883	1.898718	1.116462
##	TotInc	YrsJob	Tier_2
##	2.274076	1.229312	1.045187
##	BankSave_d	OldEmi_d	Employer_Type_Corporate
##	1.050859	1.396960	1.167121

```
model21 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d+OldEmi_d+Marital_Status,data = training_data,family=binomial)
summary(model21) ## Residual deviance: 1007.2, AIC: 1033.2 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d + Marital_Status,
##      family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   9.883e-01  6.134e-01   1.611   0.1071
## IAR           6.540e-02  7.510e-03   8.709 < 2e-16 ***
## MarVal        3.223e-07  2.426e-07   1.328   0.1840
## Term          5.723e-03  2.323e-03   2.464   0.0137 *
## FOIR         -9.535e-02  1.380e-02  -6.912 4.79e-12 ***
## LTV          -2.372e-02  5.321e-03  -4.458 8.28e-06 ***
## YrsAdd       -8.474e-03  7.429e-03  -1.141   0.2540
## TotInc       -5.750e-06  6.493e-06  -0.886   0.3758
## YrsJob       -2.134e-02  1.024e-02  -2.084   0.0372 *
## Tier_2        7.033e-01  1.755e-01   4.007 6.16e-05 ***
## BankSave_d    1.314e+00  3.223e-01   4.076 4.58e-05 ***
## OldEmi_d      1.782e+00  3.286e-01   5.424 5.83e-08 ***
## Marital_Status 1.443e-01  2.583e-01   0.559   0.5764
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1007.2  on 1243  degrees of freedom
## AIC: 1033.2
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model21) ##### more than modrate correlation
```

##	IAR	MarVal	Term	FOIR	LTV
##	3.700703	2.348767	1.303556	4.015407	1.896465
##	YrsAdd	TotInc	YrsJob	Tier_2	BankSave_d
##	1.116803	2.276321	1.141784	1.033535	1.045415
##	OldEmi_d	Marital_Status			
##	1.394603	1.045630			

```
model22 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d+OldEmi_d+Employer_Type_Govt,data = training_data,family=binomial)
summary(model22) ## Residual deviance: 1005.1, AIC: 1031.1 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d + Employer_Type_Govt,
##      family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.088e+00  5.756e-01   1.891  0.05865 .
## IAR          6.629e-02  7.558e-03   8.770 < 2e-16 ***
## MarVal       3.254e-07  2.412e-07   1.349  0.17737
## Term        6.019e-03  2.327e-03   2.587  0.00969 **
## FOIR       -9.874e-02  1.398e-02  -7.064 1.62e-12 ***
## LTV        -2.347e-02  5.310e-03  -4.420 9.89e-06 ***
## YrsAdd      -8.815e-03  7.440e-03  -1.185  0.23609
## TotInc     -5.323e-06  6.484e-06  -0.821  0.41172
## YrsJob     -2.340e-02  1.028e-02  -2.276  0.02285 *
## Tier_2      7.216e-01  1.761e-01   4.099 4.15e-05 ***
## BankSave_d   1.328e+00  3.232e-01   4.108 3.99e-05 ***
## OldEmi_d     1.836e+00  3.317e-01   5.535 3.12e-08 ***
## Employer_Type_Govt 4.584e-01  3.004e-01   1.526  0.12708
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1005.1  on 1243  degrees of freedom
## AIC: 1031.1
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model22) ##### more than modrate correlation
```

##	IAR	MarVal	Term	FOIR
##	3.735865	2.331445	1.305394	4.113588
##	LTV	YrsAdd	TotInc	YrsJob
##	1.891220	1.119030	2.263516	1.143902
##	Tier_2	BankSave_d	OldEmi_d	Employer_Type_Govt
##	1.038367	1.046489	1.411956	1.107147

```
model23 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d+OldEmi_d+Employer_Type_Govt+Build_Selfcon,data = training_data,family=binom
ial)
summary(model23) ## Residual deviance: 1005.0, AIC: 1033 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d + Employer_Type_Govt +
##      Build_Selfcon, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.080e+00  5.763e-01   1.873   0.0610 .
## IAR           6.622e-02  7.561e-03   8.759 < 2e-16 ***
## MarVal        3.286e-07  2.419e-07   1.358   0.1744
## Term          5.987e-03  2.331e-03   2.568   0.0102 *
## FOIR          -9.845e-02  1.402e-02  -7.022 2.18e-12 ***
## LTV           -2.359e-02  5.332e-03  -4.425 9.65e-06 ***
## YrsAdd        -8.907e-03  7.450e-03  -1.196   0.2318
## TotInc        -5.124e-06  6.522e-06  -0.786   0.4321
## YrsJob        -2.330e-02  1.029e-02  -2.265   0.0235 *
## Tier_2         7.255e-01  1.767e-01   4.106 4.02e-05 ***
## BankSave_d     1.313e+00  3.273e-01   4.012 6.02e-05 ***
## OldEmi_d       1.830e+00  3.321e-01   5.511 3.56e-08 ***
## Employer_Type_Govt 4.619e-01  3.007e-01   1.536   0.1245
## Build_Selfcon  1.156e-01  4.445e-01   0.260   0.7948
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1005.0  on 1242  degrees of freedom
## AIC: 1033
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model23) ##### more than modrate correlation
```

```
##              IAR              MarVal              Term              FOIR
##      3.739391          2.334744          1.310028          4.138480
##              LTV              YrsAdd              TotInc              YrsJob
##      1.907479          1.121688          2.291303          1.145366
##              Tier_2          BankSave_d          OldEmi_d Employer_Type_Govt
##      1.045801          1.075939          1.417117          1.109268
##      Build_Selfcon
##      1.126577
```

```
model24 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d+OldEmi_d+Employer_Type_Govt+Tier_3,data = training_data,family=binomial)
summary(model24) ## Residual deviance: 1004.7, AIC: 1032.7 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d + Employer_Type_Govt +
##      Tier_3, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.183e+00  5.962e-01   1.984   0.0473 *
## IAR           6.574e-02  7.611e-03   8.638  < 2e-16 ***
## MarVal        3.281e-07  2.415e-07   1.359   0.1743
## Term          5.863e-03  2.342e-03   2.503   0.0123 *
## FOIR         -9.786e-02  1.406e-02  -6.961  3.38e-12 ***
## LTV          -2.356e-02  5.314e-03  -4.433  9.29e-06 ***
## YrsAdd        -9.327e-03  7.488e-03  -1.246   0.2129
## TotInc        -5.200e-06  6.478e-06  -0.803   0.4221
## YrsJob        -2.339e-02  1.029e-02  -2.274   0.0230 *
## Tier_2         6.553e-01  2.064e-01   3.175   0.0015 **
## BankSave_d     1.325e+00  3.227e-01   4.107  4.00e-05 ***
## OldEmi_d       1.822e+00  3.323e-01   5.482  4.21e-08 ***
## Employer_Type_Govt 4.668e-01  3.012e-01   1.550   0.1212
## Tier_3        -1.205e-01  1.943e-01  -0.620   0.5353
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1004.7  on 1242  degrees of freedom
## AIC: 1032.7
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model24) ##### more than modrate correlation
```

```
##              IAR              MarVal              Term              FOIR
##      3.788591          2.331415          1.320158          4.158073
##              LTV              YrsAdd              TotInc              YrsJob
##      1.892252          1.133962          2.262764          1.144198
##              Tier_2          BankSave_d          OldEmi_d Employer_Type_Govt
##      1.427530          1.046890          1.415700          1.110003
##              Tier_3
##      1.433995
```

```
model25 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d+OldEmi_d+Employer_Type_Govt+Accommodation_Class,data = training_data,family
=binomial)
summary(model25) ## Residual deviance: 1004.4, AIC: 1032.4 ##
```



```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d + Employer_Type_Govt +
##      Accommodation_Class, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.220e+00  5.981e-01   2.040   0.0413 *
## IAR             6.520e-02  7.654e-03   8.518 < 2e-16 ***
## MarVal          3.215e-07  2.408e-07   1.335   0.1818
## Term            5.803e-03  2.340e-03   2.480   0.0131 *
## FOIR           -9.724e-02  1.406e-02  -6.916 4.66e-12 ***
## LTV            -2.384e-02  5.349e-03  -4.458 8.29e-06 ***
## YrsAdd          -6.589e-03  7.918e-03  -0.832   0.4053
## TotInc          -5.546e-06  6.515e-06  -0.851   0.3946
## YrsJob          -2.284e-02  1.030e-02  -2.217   0.0266 *
## Tier_2           7.112e-01  1.765e-01   4.029 5.59e-05 ***
## BankSave_d       1.316e+00  3.237e-01   4.066 4.78e-05 ***
## OldEmi_d         1.816e+00  3.325e-01   5.462 4.70e-08 ***
## Employer_Type_Govt 4.584e-01  3.003e-01   1.527   0.1268
## Accommodation_Class -1.483e-01  1.806e-01  -0.821   0.4114
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1004.4  on 1242  degrees of freedom
## AIC: 1032.4
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model25) ##### more than modrate correlation
```

```
##              IAR              MarVal              Term              FOIR
##      3.825827          2.334029          1.318419          4.159156
##              LTV              YrsAdd              TotInc              YrsJob
##      1.909242          1.263887          2.277752          1.147229
##              Tier_2          BankSave_d          OldEmi_d Employer_Type_Govt
##      1.042718          1.047462          1.417796          1.106595
## Accommodation_Class
##      1.283060
```

```
model26 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d+OldEmi_d+Employer_Type_Govt+Employer_Type_Business,data = training_data,fam
ily=binomial)
summary(model26) ## Residual deviance: 997.67, AIC: 1025.7 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d + Employer_Type_Govt +
##      Employer_Type_Business, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.176e+00  5.776e-01   2.037  0.04168 *
## IAR             6.713e-02  7.591e-03   8.843 < 2e-16 ***
## MarVal          3.657e-07  2.424e-07   1.509  0.13132
## Term            6.518e-03  2.340e-03   2.785  0.00535 **
## FOIR           -1.038e-01  1.421e-02  -7.307 2.72e-13 ***
## LTV            -2.262e-02  5.292e-03  -4.275 1.91e-05 ***
## YrsAdd          -8.837e-03  7.470e-03  -1.183  0.23686
## TotInc          -3.570e-06  6.463e-06  -0.552  0.58066
## YrsJob          -1.086e-02  1.137e-02  -0.955  0.33947
## Tier_2           7.054e-01  1.769e-01   3.989 6.65e-05 ***
## BankSave_d       1.306e+00  3.232e-01   4.040 5.36e-05 ***
## OldEmi_d         1.896e+00  3.333e-01   5.688 1.29e-08 ***
## Employer_Type_Govt  1.115e-01  3.267e-01   0.341  0.73287
## Employer_Type_Business -5.331e-01  1.974e-01  -2.701  0.00692 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.23  on 1255  degrees of freedom
## Residual deviance:  997.67  on 1242  degrees of freedom
## AIC: 1025.7
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model26) ##### more than modrate correlation
```

```
##              IAR              MarVal              Term
##      3.762913          2.320422          1.315722
##      FOIR              LTV              YrsAdd
##      4.225549          1.890085          1.119733
##      TotInc            YrsJob            Tier_2
##      2.254816          1.356108          1.038444
##      BankSave_d        OldEmi_d        Employer_Type_Govt
##      1.048909          1.417689          1.312345
## Employer_Type_Business
##      1.514230
```

###	IAR	MarVal	Term	FOIR	LTV	YrsAdd
TotInc	YrsJob	Tier_2	BankSave_d	OldEmi_d	Emplo	
yer_Type_Govt	Employer_Type_Business					
## 3.762913	2.320422	1.315722	4.225549	1.890085	1.11973	
2.254816	1.356108	1.038444	1.048909	1.417689		
1.312345	1.514230					

```

model27 = glm(Decision ~ IAR+MarVal+Term+FOIR+LTV+YrsAdd+TotInc+YrsJob+Tier_2+Bank
Save_d+OldEmi_d+Employer_Type_Govt+Employer_Type_Business+Tier_1,data = training_d
ata,family=binomial)
summary(model27) ## Residual deviance: 997.38, AIC: 1027.4 ##

```

```

##
## Call:
## glm(formula = Decision ~ IAR + MarVal + Term + FOIR + LTV + YrsAdd +
##      TotInc + YrsJob + Tier_2 + BankSave_d + OldEmi_d + Employer_Type_Govt +
##      Employer_Type_Business + Tier_1, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.154e+00  5.795e-01   1.992 0.046402 *
## IAR             6.663e-02  7.646e-03   8.714 < 2e-16 ***
## MarVal          3.681e-07  2.427e-07   1.517 0.129342
## Term            6.377e-03  2.357e-03   2.706 0.006809 **
## FOIR            -1.031e-01  1.429e-02  -7.210 5.59e-13 ***
## LTV             -2.271e-02  5.297e-03  -4.287 1.81e-05 ***
## YrsAdd          -9.297e-03  7.521e-03  -1.236 0.216436
## TotInc          -3.475e-06  6.459e-06  -0.538 0.590595
## YrsJob          -1.092e-02  1.137e-02  -0.960 0.336895
## Tier_2           7.530e-01  1.973e-01   3.816 0.000136 ***
## BankSave_d       1.303e+00  3.228e-01   4.035 5.45e-05 ***
## OldEmi_d         1.882e+00  3.341e-01   5.634 1.77e-08 ***
## Employer_Type_Govt 1.215e-01  3.275e-01   0.371 0.710746
## Employer_Type_Business -5.299e-01  1.974e-01  -2.684 0.007275 **
## Tier_1           1.053e-01  1.954e-01   0.539 0.590117
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.23  on 1255  degrees of freedom
## Residual deviance:  997.38  on 1241  degrees of freedom
## AIC: 1027.4
##
## Number of Fisher Scoring iterations: 5

```

```
vif(model27) ##### more than modrate correlation
```

##	IAR	MarVal	Term
##	3.818901	2.321083	1.331724
##	FOIR	LTV	YrsAdd
##	4.272216	1.891563	1.135421
##	TotInc	YrsJob	Tier_2
##	2.254656	1.356480	1.293599
##	BankSave_d	OldEmi_d	Employer_Type_Govt
##	1.049410	1.423660	1.315699
##	Employer_Type_Business	Tier_1	
##	1.514802	1.318157	

```
## Model with maximum significant is model26 ##
```

```
### Prediction
```

```
res = predict(model26,testing_data,type="response")
head(res)
```

##	3	13	16	17	25	27
##	0.8880594	0.6102681	0.9958791	0.7820962	0.9830764	0.8956050

```
head(testing_data$Decision)
```

```
## [1] 1 1 1 1 1 1
```

```
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
pred_class = ifelse (res >0.75,1,0)
```

```
confusionMatrix(as.factor(pred_class),as.factor(testing_data$Decision),positive =
"1")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0  41  49
##           1  22 202
##
##           Accuracy : 0.7739
##           95% CI : (0.7236, 0.819)
##           No Information Rate : 0.7994
##           P-Value [Acc > NIR] : 0.883444
##
##           Kappa : 0.3926
##
## Mcnemar's Test P-Value : 0.002031
##
##           Sensitivity : 0.8048
##           Specificity : 0.6508
##           Pos Pred Value : 0.9018
##           Neg Pred Value : 0.4556
##           Prevalence : 0.7994
##           Detection Rate : 0.6433
##           Detection Prevalence : 0.7134
##           Balanced Accuracy : 0.7278
##
##           'Positive' Class : 1
##
```

```
## Accuracy : 0.7739, Sensitivity : 0.8048 ##
```

```
##           Reference
## Prediction    0    1
##           0  41  49
##           1  22 202
```

```
model28 = glm(Decision ~ IAR+Term+FOIR+LTV+YrsAdd+TotInc+Tier_2+BankSave_d+OldEmi_
d+Employer_Type_Business,data = training_data,family=binomial)
summary(model28) ## Residual deviance: 1001.0, AIC: 1023 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + Term + FOIR + LTV + YrsAdd + TotInc +
##      Tier_2 + BankSave_d + OldEmi_d + Employer_Type_Business,
##      family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.182e+00  5.581e-01   2.118 0.034169 *
## IAR             6.697e-02  7.578e-03   8.838 < 2e-16 ***
## Term           7.973e-03  2.171e-03   3.673 0.000240 ***
## FOIR          -9.631e-02  1.326e-02  -7.264 3.77e-13 ***
## LTV           -2.813e-02  4.084e-03  -6.888 5.65e-12 ***
## YrsAdd        -9.749e-03  7.306e-03  -1.334 0.182094
## TotInc         2.088e-06  5.331e-06   0.392 0.695268
## Tier_2         7.038e-01  1.758e-01   4.002 6.27e-05 ***
## BankSave_d     1.262e+00  3.207e-01   3.937 8.26e-05 ***
## OldEmi_d       1.759e+00  3.199e-01   5.500 3.80e-08 ***
## Employer_Type_Business -5.898e-01  1.705e-01  -3.459 0.000543 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1001.0  on 1245  degrees of freedom
## AIC: 1023
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model28) ##### more than modrate correlation
```

```
##              IAR              Term              FOIR
##      3.729437      1.143658      3.665973
##              LTV              YrsAdd              TotInc
##      1.105130      1.071669      1.545327
##              Tier_2      BankSave_d      OldEmi_d
##      1.030638      1.035556      1.325454
## Employer_Type_Business
##      1.131814
```

```
model29 = glm(Decision ~ IAR+Term+FOIR+LTV+YrsAdd+Tier_2+BankSave_d+OldEmi_d+Employer_Type_Business,data = training_data,family=binomial)
summary(model29) ## Residual deviance: 1001.2, AIC: 1021.2 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + Term + FOIR + LTV + YrsAdd + Tier_2 +
##       BankSave_d + OldEmi_d + Employer_Type_Business, family = binomial,
##       data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.300523    0.470693   2.763 0.005727 **
## IAR              0.066066    0.007204   9.170 < 2e-16 ***
## Term            0.007975    0.002171   3.674 0.000239 ***
## FOIR           -0.096527    0.013246  -7.287 3.16e-13 ***
## LTV            -0.028043    0.004068  -6.894 5.42e-12 ***
## YrsAdd         -0.010036    0.007269  -1.381 0.167351
## Tier_2          0.701654    0.175824   3.991 6.59e-05 ***
## BankSave_d       1.262910    0.320465   3.941 8.12e-05 ***
## OldEmi_d         1.778601    0.316518   5.619 1.92e-08 ***
## Employer_Type_Business -0.578913    0.168269  -3.440 0.000581 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1001.2  on 1246  degrees of freedom
## AIC: 1021.2
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model29) ##### more than modrate correlation
```

##	IAR	Term	FOIR
##	3.382017	1.142342	3.664120
##	LTV	YrsAdd	Tier_2
##	1.100501	1.060931	1.029790
##	BankSave_d	OldEmi_d	Employer_Type_Business
##	1.035834	1.294511	1.102311

```
model30 = glm(Decision ~ IAR+Term+FOIR+LTV+Tier_2+BankSave_d+OldEmi_d+Employer_Typ
e_Business,data = training_data,family=binomial)
summary(model30) ## Residual deviance: 1003.0, AIC: 1021 ##
```

```
##
## Call:
## glm(formula = Decision ~ IAR + Term + FOIR + LTV + Tier_2 + BankSave_d +
##      OldEmi_d + Employer_Type_Business, family = binomial, data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.111295    0.449048   2.475 0.013332 *
## IAR              0.067081    0.007176   9.348 < 2e-16 ***
## Term            0.008584    0.002125   4.039 5.38e-05 ***
## FOIR           -0.098433    0.013202  -7.456 8.93e-14 ***
## LTV            -0.028064    0.004046  -6.936 4.02e-12 ***
## Tier_2          0.701300    0.175712   3.991 6.57e-05 ***
## BankSave_d       1.246949    0.319093   3.908 9.31e-05 ***
## OldEmi_d         1.777726    0.316214   5.622 1.89e-08 ***
## Employer_Type_Business -0.587475    0.168072  -3.495 0.000473 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1259.2  on 1255  degrees of freedom
## Residual deviance: 1003.0  on 1247  degrees of freedom
## AIC: 1021
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model30) ##### more than modrate correlation
```

```
##              IAR              Term              FOIR
##      3.368579      1.095716      3.646433
##              LTV              Tier_2      BankSave_d
##      1.100407      1.030081      1.034882
##      OldEmi_d Employer_Type_Business
##      1.296489      1.101963
```

```
res = predict(model30,testing_data,type="response")
head(res)
```

```
##           3           13           16           17           25           27
## 0.8234850 0.6206316 0.9908924 0.7803497 0.9850489 0.9067153
```

```
head(testing_data$Decision)
```

```
## [1] 1 1 1 1 1 1
```



```
library(caret)
pred_class = ifelse (res >0.75,1,0)

confusionMatrix(as.factor(pred_class),as.factor(testing_data$Decision),positive =
"1")
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction    0    1
##              0  40  52
##              1  23 199
##
##              Accuracy : 0.7611
##              95% CI : (0.7101, 0.8072)
##              No Information Rate : 0.7994
##              P-Value [Acc > NIR] : 0.958620
##
##              Kappa : 0.3649
##
## Mcnemar's Test P-Value : 0.001224
##
##              Sensitivity : 0.7928
##              Specificity : 0.6349
##              Pos Pred Value : 0.8964
##              Neg Pred Value : 0.4348
##              Prevalence : 0.7994
##              Detection Rate : 0.6338
##              Detection Prevalence : 0.7070
##              Balanced Accuracy : 0.7139
##
##              'Positive' Class : 1
##
```

```
## Accuracy : 0.7611, Sensitivity : 0.7928 ##
##              Reference
## Prediction    0    1
##              0  40  52
##              1  23 199
```